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Production of a Catalan-Specific Vowel Contrast by Early Spanish-Catalan Bilinguals

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Abstract

The present study investigates the acoustics (F1 × F2) of Catalan and Spanish mid-back vowels as produced by highly proficient, early Spanish-Catalan bilinguals residing on the island of Majorca, a bilingual speech community. Majorcan Catalan has two phonemic mid-back vowels in stressed positions (/o/ and /ɔ/) while Spanish possesses only one (/o/). Two groups of bilinguals were recruited and asked to produce materials in both languages – one group of Spanish-dominant and one of Catalan-dominant speakers. It was first found that Catalan and Spanish /o/ are virtually indistinguishable. Catalan /ɔ/ is lower and more fronted than the other two vowels. Spanish-dominant bilinguals were found to differ from Catalan-dominant ones in that they did not produce the Catalan-specific /o/-/ɔ/ contrast in their speech; that is, they produced a single, merged Catalan mid-back vowel. A within-subjects analysis of first- and second-language mid-back vowels further suggested, for Spanish-dominant bilinguals, that they had developed a separate vowel category to accommodate their single, merged Catalan mid-back vowel; that is, they possessed a two-category mid-back vowel system, i.e. one for their Spanish /o/ and one for their merged Catalan /o/ + /ɔ/. Potential explanations and theoretical implications are discussed.

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Introduction

Many studies have shown that learning the sounds of a non-native language in a nativelike manner is a challenging task. Learners experience difficulties with non-native sounds even after years of experience with the language [Flege et al., 1995; Oyama, 1976; Piske et al., 2001; among others]. These difficulties may be observed in production [Flege et al., 1999a, b; Yeni-Komshian et al., 2000; among others] as well as in perception [Pallier et al., 1997; Sebastián-Gallés and Soto-Faraco, 1999; among others].

It is by now an established finding that earlier exposure is 'better' than late exposure with regards to nativelike attainment in speech learning [Flege, 1999, 2003, 2007; Oyama, 1976; Yeni-Komshian et al., 2000]. A lag of even a few years between the age

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of learning of the L1 and that of the L2 is generally thought to be sufficient to trigger difficulties regarding nativelike attainment of the L2 [Pallier et al., 1997; Sebastián-Gallés, 1999]. It has been shown that children become 'attuned' to their native language very early in life [Bosch and Sebastián-Gallés, 2003; Kuhl et al., 1992, 2006; Werker and Tees, 1984]. The adaptation of the speech performance system to the characteristics of the L1 is thought to facilitate L1 development as well as to hinder future L2 learning [see Kuhl et al., 2008, for a review].

This study is concerned with the production of the Catalan-specific /o/-/ɔ/ contrast by Spanish-Catalan early bilinguals. The perception abilities, including those applicable to Catalan-specific vowel contrasts, of a population very similar to the one investigated here have been analyzed in a number of recent papers. In a study of perceptual identification and discrimination of a seven-step /e/-/ɛ/ continuum, Pallier et al. [1997] found robust differences between Spanish- and Catalan-dominant Spanish-Catalan bilinguals in Barcelona regarding the perception of this Catalan-specific vowel contrast. Many, but not all, Spanish-dominant subjects basically displayed flat responses in both the identification and discrimination experiments [see also Bosch et al., 2000]. Using an adaptation of the gating task that involved a two-alternative forced choice, Sebastián-Gallés and Soto-Faraco [1999] investigated four Catalan-specific contrasts, i.e. /o/-/ɔ/, /e/-/ɛ/, /s/-/z/, /s/-/ʃ/. Sebastián-Gallés and Soto-Faraco [1999] showed that even Spanish-dominant bilinguals who displayed nativelike (i.e., Catalan-dominantlike) identification when presented, for instance, with full /e/-/ɛ/ vowels as stimuli displayed processing patterns that made them different from Catalan-dominant subjects under the conditions of a gating task. In other words, while some Spanish-dominant bilinguals might be able to correctly discriminate between Catalan /e/ and /ɛ/ in off-line tasks, most are revealed not to be nativelike in online tasks. An off-line task is one in which the researcher has access only to the final decision/response of a subject while in online tasks the researcher is able to observe the decision-/response-making process of the subject. For instance, categorical responses to trials in perceptual identification and discrimination experiments provide off-line data while response latencies (i.e., reaction times) in these experiment designs open a window to the decision-making process and thus provide online data. Gating tasks also provide online data since they open a window to the decision-making process of the subject as it unfolds. Pallier et al. [2001] showed that non-native perception difficulties have consequences for lexical access. These authors found that Spanish-dominant bilinguals presented repetition priming for minimal pairs exhibiting the /e/-/ɛ/ contrast while Catalan-dominant subjects did not. This indicated that most Spanish-dominant listeners processed Catalan minimal pairs as homophones, unlike Catalan-dominant listeners. Navarra et al. [2005] asked a group of Spanish-Catalan bilinguals to categorize the first syllable of disyllabic stimuli in words that could exhibit an /e/-/ɛ/ variation on the second, task-irrelevant, syllable. Catalan-dominant subjects were slowed down by the presence of /e/-/ɛ/ differences on the second syllable, while Spanish-dominant participants were not, further suggesting that they were not able to process the Catalan-specific /e/-/ɛ/ contrast in nativelike manner. Finally, Sebastián-Gallés et al. [2005] extended these studies to investigate speech perception in simultaneous bilinguals, i.e. subjects who grew up in bilingual homes. It was found that, even for these subjects, one dominant language prevailed; that is, they behaved as either Catalan- or Spanish-dominant early bilinguals.

The perception studies reviewed above show, for early Spanish-Catalan bilinguals residing in Barcelona, that highly proficient Spanish-dominant subjects tend to differ

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from Catalan-dominant subjects in their processing of Catalan-specific phonemic contrasts. A relevant finding was that even though some Spanish-dominant participants were nativelike (i.e., Catalan-dominant-like) when provided with full-vowel stimuli (i.e., in off-line tasks), they differed from Catalan-dominant subjects when the experimental conditions allowed the researchers to investigate the decision-making process (i.e., in online tasks). In other words, online tasks, which are arguably more sensitive than off-line ones, tend to reveal processing differences between native speakers and even the most highly proficient early bilinguals. This finding might have consequences for bilingual speech production. While much is known about the perceptual patterns of Spanish-Catalan bilinguals in Barcelona, their production of Catalan or Spanish sounds largely remains to be investigated [cf. Cortés et al., 2009; Mora and Keidel, 2010; Mora and Nadeu, 2009]. Are Spanish-dominant early bilinguals likely to become nativelike (i.e., indistinguishable from Catalan-dominant subjects) in their production? It is possible that some Spanish-dominant speakers are able to produce Catalan-specific vowel contrasts in a nativelike manner, as some were found to able to perceive Catalan-specific contrasts in off-line tasks. Also, may their production of Catalan and Spanish sounds reveal complex interactions between the two phonetic (sub)systems that perception data may not be able to reveal? Perception tests have tended to focus on the discriminability and identifiability of L2-specific vowel contrasts. However, very little is known about the relationship between native and nonnative phonetic categories for these bilinguals. If they are not able to produce the L2-specific contrast, does that mean that they use a merged L1 + L2 category? A merged L1 + L2 category is a phonetic category that displays acoustic characteristics of both an L1 and an L2 sound. On the other hand, if they produce the L2-specific contrast, do they do so in a nativelike manner?

While perception studies have tended to find robust differences between early bilinguals and monolingual speakers of the L2, production data of early bilinguals shows a less clear-cut picture [Piske et al., 2002]. On the one hand, it has been shown that early bilinguals might be able to attain nativelike pronunciation, in the sense that they might be functionally indistinguishable from monolingual speakers of the L2 with regard to their pronunciation of certain sounds. For instance, Munro et al. [1996] asked native, monolingual English speakers to rate (on a 1 = 'wrong' to 5 = 'correct' continuum) the English vowels produced by a group of Italian-English bilinguals. It was found that many early bilinguals produced English vowels in a way indistinguishable from monolingual English speakers. Flege et al. [1999a] investigated two groups of early Italian-English bilinguals, one that self-reportedly used Italian frequently and one that used it seldom. The authors asked monolingual English listeners to identify the English vowels produced by the bilinguals. None of the two groups of early bilinguals differed from a group of native English speakers in their productions.

On the other hand, differences between monolingual speakers and early bilinguals have been found in some studies. For instance, Mack [1989] explored the English productions of early French-English bilinguals and found that these bilinguals differed from English monolinguals in producing vowel /i/. Guion et al. [2000] examined the productions of three groups of early Quichua-Spanish bilinguals through a study of 'foreign' accent ratings. The three groups of early bilinguals differed in their amount of Quichua use. It was found that the Spanish productions of bilinguals who used Quichua frequently were more 'accented' than those of monolingual Spanish speakers, while bilinguals who used Quichua seldom did not differ from Spanish monolinguals in their

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productions. Piske et al. [2002] reexamined the productions of the speakers studied in Flege et al. [1999a] with a different analytical procedure. Native English listeners were asked to rate vowels produced by bilinguals. It was found that some of the vowels of the early learners who used Italian frequently received lower ratings than those produced by native English subjects while none of those produced by the learners who used Italian seldom differed from the native controls.

Available evidence leads to the hypothesis that Spanish-dominant subjects will experience more difficulties in producing Catalan vowels than Catalan-dominant speakers will experience in producing Spanish vowels. It appears that developing an L2-specific sound contrast that does not have an equivalent in the L1, i.e. when the two L2 sounds are assimilated to a single L1 sound, is especially challenging for language learners [Best, 1995]. We hypothesize that this is the situation encountered by Spanish speakers who learn Catalan sequentially. Notice, however, that no explicit perceptual assimilation studies have been conducted for this specific Catalan vowel contrast. On the other hand, how nativelike will the Spanish mid-back vowel productions of the Catalan-dominant bilinguals be? Escudero and Boersma [2002] discuss some evidence suggesting that some bilinguals may develop a spurious L2 contrast when two L1 sounds share the acoustic-perceptual space occupied by a single L2 sound. A spurious L2 contrast is a nonauthentic, nongenuine contrast; that is, a spurious contrast is a contrast that does not exist in the productions of native speakers of the L2 but that may be nongenuinely added by non-native speakers. Since this may be the case for Spanish /o/ and Catalan /o/ and /ɔ/, we may ask: do Catalan-dominant bilinguals produce a spurious mid-back vowel contrast in their Spanish (at least in cognates)?

As mentioned above, Catalan has two phonemic mid-back vowels (/o/ and /ɔ/), whereas Spanish has only one (/o/). As far as we know, no acoustic studies have explicitly compared Catalan and Spanish mid-back vowels. Recasens and Espinosa [2006, 2009] recently carried out detailed acoustic investigations of the vowel systems of several Catalan dialects. Specifically for Majorcan Catalan, the dialect that concerns us here, Recasens and Espinosa [2006] showed, for 5 male speakers, that the /o/-/ɔ/ contrast was robust and manifested in both F1 and F2. Vowel charts suggest that the magnitude of the acoustic difference between /o/ and /ɔ/ is much larger for Majorcan than for other dialects of Catalan, such as Eastern, Western and Valencian [Recasens and Espinosa, 2006, p. 654]. Furthermore, these authors showed that Majorcan Catalan /ɔ/ is especially low, almost as much as /a/. Average values show a mean difference of 161 Hz for F1 (/ɔ/ = 708 Hz; /o/ = 547 Hz) and 178 Hz for F2 (/ɔ/ = 1,178 Hz; /o/ = 1,000 Hz) between /o/ and /ɔ/ [Recasens and Espinosa, 2006, p. 655]. Data reported in Martínez Celdrán [1984] yield the following average values for Spanish /o/: 506 Hz for F1 and 1,037 Hz for F2. Bradlow [1995] reports the following mean values for Spanish /o/: 460 Hz for F1 and 1,019 Hz for F2 [see also Martínez Celdrán, 1995; Quilis and Esgueva, 1983]. According to the values reported above, Spanish /o/ is somewhere in between Catalan /o/ and /ɔ/ [Sebastián-Gallés and Soto-Faraco, 1999]. The values reported in previous studies suggest a high degree of similarity between Spanish /o/ and Catalan /o/, with a slight difference in height between these two vowels. Apparently, Spanish /o/ and Catalan /ɔ/ display a larger acoustic distance. In other words, whereas it seems that Catalan /ɔ/ and Spanish /o/ are robustly different in their acoustic characteristics, it remains to be seen whether Catalan /o/ and Spanish /o/ are statistically different or not. In yet

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other words, it remains to be seen whether Catalan /o/ productions by native Catalan speakers are indeed acoustically different from Spanish /o/ productions by Spanish native speakers.

The present study is organized as follows. First, the Spanish mid-back vowels of the Spanish-dominant participants were compared with the Catalan mid-back vowels of the Catalan-dominant participants. This was done to investigate the relationship (i.e., relative acoustic distance) between Spanish /o/ and Catalan /o/ and /ɔ/. Second, we compared the vowels of Catalan and then of Spanish as produced by dominant and nondominant speakers of each language. That is, we first compared the Catalan vowels produced by the Catalan-dominant subjects with the Catalan vowels produced by the Spanish-dominant bilinguals and then compared the Spanish vowels produced by the Spanish-dominant subjects with the Spanish vowels produced by the Catalan-dominant speakers. The question we addressed was as follows: Are nondominant productions of early bilinguals nativelike? Third, the interaction (within-subject) patterns between the dominant and nondominant phonetic (sub)systems of the bilinguals were analyzed. This was done mostly in order to investigate whether Spanish-dominant bilinguals had been able to develop two different acoustic distributions for the two Catalan mid-back vowel phonemes and/or whether they had developed three different vowel categories, one for Spanish /o/, one for Catalan /o/ and one for Catalan /ɔ/. Additionally, mostly for the sake of completeness, the Catalan and Spanish productions of the Catalan-dominant participants were compared. Since we used Catalan-Spanish cognates in our experimental materials, it could be hypothesized that Catalan-dominant speakers might transfer their dominant phonetic (sub) system to their nondominant language and thus implement a spurious phonetic contrast between two Spanish vowels.

Method

Participants

Twenty Spanish-Catalan bilinguals were recruited for their participation in a production study. The bilingual speakers were classified in two groups according to their dominant language. Classification was based on two criteria, a language background questionnaire and the results of an accent-rating task. Balanced numbers of males and females were recruited for both groups. All of the participants were born and raised in Majorca. They were born during the decade of the 1970s or early 1980s. At the time of the recordings, they had all attained a secondary-education degree and some were pursuing university education. Their education, during childhood, was administered in both Catalan and Spanish. All of the participants were thus literate and proficient in both languages at the time of testing.

A language-background questionnaire was administered in the language of choice of the participants. The questions were asked prior to their participation in the experiment, at the beginning of the session. It included the following questions: (1) Which language do you consider to be your native language? (2) Which language did you use to speak at home, as a child, with your family? (3) Which language do you use more frequently in your daily life? (4) Rate the percentage of use of Spanish/Catalan in your daily life: (i) with family, (ii) with friends, (iii) at work/school, (iv) while shopping, (v) with strangers. Subjects were classified as Catalan-dominant if they claimed Catalan to be their native language, used Catalan exclusively with their parents and siblings and self-reportedly use Catalan more frequently than Spanish in their daily lives. Additionally, higher percentages of Catalan than of Spanish use needed to be reported for the different conditions in question 4. On the other hand, subjects were classified as Spanish-dominant if they claimed Spanish to be their native language, used

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Spanish exclusively with their parents and siblings and self-reportedly use Spanish more frequently than Catalan in their daily lives. Individuals who used both languages at home, with their parents or siblings, or reported similar percentages of use of Catalan and Spanish were excluded. Thus, although all the participants are early or childhood bilinguals, they may be claimed to have one dominant and one nondominant language at least based on external factors.

An accent-rating task was designed in order to verify, with a speech proficiency measure, the classification of the subjects in the two dominance groups. Twenty Catalan-Spanish bilingual raters, 10 Catalan-dominant and 10 Spanish dominant (5 males and 5 females in each group), were recruited using the same procedure described above for the bilingual speakers. The bilingual speakers were recorded while reading materials in both their dominant and their nondominant language. The raters listened to and rated speech materials only in their dominant language. Therefore, Catalan-dominant raters listened to Catalan materials that had been recorded by Catalan-dominant and Spanish-dominant speakers, and Spanish-dominant raters listened to Spanish materials recorded by Spanish-dominant and Catalan-dominant speakers. The results confirmed the initial classification based on the language background questionnaire, i.e. Catalan-dominant productions were robustly identified as native Catalan and non-native Spanish productions, also Spanish-dominant productions were systematically identified as native Spanish and non-native Catalan productions [see Simonet, 2010a, b, for details].

Recordings

Eight one-, two- or three-syllable Catalan words with Spanish cognates were selected, four with /o/ in its stressed syllable and four with /ɔ/. The Catalan /o/-/ɔ/ contrast is implemented only in stressed syllables. In Majorcan Catalan, in particular, /o/ and /ɔ/ are neutralized to /o/ when unstressed. The materials were adapted from some of those used in Recasens and Espinosa [2006, 2009]. The Catalan words were then translated to Spanish, using cognates. This resulted in four Spanish words with Catalan /o/ cognates and four words with Catalan /ɔ/ cognates. Following conventions in Catalan linguistics, /o/ and /ɔ/ are referred to here as the *closed* and *open* mid-back vowel phonemes, respectively. Thus, for the sake of orthogonality, the Spanish cognates are also classified into two groups, *closed* and *open* mid-back vowels. The eight Catalan words and the eight Spanish cognates resulted in a total of 16 target lexical items to be produced by the bilingual subjects. Mid-back vowels were surrounded by bilabial consonants in half of the instances and by dentoalveolars in the other half (see table 1 for a list of the materials).

The participants were asked to read aloud two lists of sentences, one per language, which were provided to them on paper. The bilingual participants read first the materials in their dominant language and, after participating in an interactive game with the author in their native language and taking a short break, they read the materials in their nondominant language. Each language list contained five sublists, that is, five blocked pseudorandomizations of the target list. Each of the sublists contained the eight target words, which were pseudorandomly interspersed among many (n = 51) distracters. These distracters were used as materials for other studies [Simonet, 2010a, b].

A total of 1,600 tokens, 800 in Spanish and 800 in Catalan, were recorded. The total data set had the following structure: 2 (native language groups) × 10 (participants per native language group) × 2 (languages) × 8 (words per language) × 5 (iterations per word) = 1,600 items. A total of 12 tokens were excluded due to recording errors. Therefore, the dataset comprised a total of 1,588 tokens, 793 recorded by Catalan-dominant and 795 recorded by Spanish-dominant subjects.

The materials were recorded in a quiet room, in the home or workplace of the speakers, using a solid-state digital recorder (Marantz PMD660) and a head-worn dynamic microphone (Shure SM10A). Speech was digitized at 44.1 kHz (16-bit quantization) and later down-sampled at 22.05 kHz.

Acoustic Analyses

Vowels were segmented with Praat [Boersma, 2001], using standard procedures. Synchronized waveform and spectrographic displays were used for visual segmentation. Formant trajectories, especially the second formant (F2), as well as intensity displays were taken as indicators of vowel onsets and offsets. The onset of a vowel was marked on the first pitch period in which F2 was visible and/or

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Table 1. Catalan and Spanish materials

Language	Vowel	Sentence	Translation
Catalan	/o/	No tolera sa lactosa	She is lactose-intolerant
		S'ha banyada tota	She got all wet
		Agafa-ho des pom	Hold it from the doorknob
		Sa proa i sa popa	The bow and the stern
	/ɔ/	Ell la destrossa	He destroys it
		Dona-me'n un tros	Give me a piece
		Telèfon mòbil	Mobile phone
		Me causa fòbia	It terrifies me
Spanish	/o/	No tolera la lactosa	She is lactose-intolerant
		Se mojó toda	She got all wet
		No toques el pomo	Do not touch the doorknob
		La proa y la popa	The bow and the stern
	'/ɔ/'	Él la destroza	He destroys it
		Dame un trozo	Give me a piece
		Teléfono móvil	Mobile phone
		Me causa fobia	It terrifies me

an intensity similar to that of the vowel's midpoint was present. In the case of spirantized surrounding consonants, an intensity dip was always observable in our data and thus we were able to use the procedure outlined in the previous sentence. The offset was marked on the last such pitch period. First (F1) and second (F2) formant values were automatically extracted, using Praat, from the vowel's midpoint, which was mathematically determined. Formant tracks were calculated with the Burg algorithm as implemented in Praat. The effective window length for the calculation was set at 25 ms, which is the Praat standard. This window length was maintained across tokens and speakers. The maximum number of formants to be located by the formant tracker in Praat was always 5 and the ceiling was set at 5.0 kHz for males and 5.5 kHz for females, which are the Praat standard parameters and were deemed appropriate after visual inspection of all sound files. Formant values were extracted in Hertz and then further normalized following a speaker-intrinsic procedure.

An investigation into the speakers' vowel spaces showed the need for a by-subject normalization of F1 and F2. By-subject normalization was thought to be necessary since speakers were found to vary in the location and width of their acoustic vowel (F1 × F2) spaces, as evidenced by mean F1 and F2 values for these subjects' point vowels /i a u/. The most robust normalization methods, i.e. those that retain structural or social-indexical differences while minimizing differences due to anatomy, including vocal tract size, are generally speaker- and formant-intrinsic but vowel-extrinsic [Adank et al., 2004; Fabricius et al., 2009]. The selected speaker-intrinsic normalization procedure took into account the vowel spaces of the speakers, as determined by their point vowels. The procedure was based on the Lobanov method [Adank et al., 2004; Lobanov, 1971; Nearey, 1977]. The Lobanov method has the following formula:

$$F[norm] = (F[Hz] - M[Hz]) / SD[Hz]$$

where F[norm] is the normalized formant value, F[Hz] is the target formant value in Hertz that one wishes to normalize, M[Hz] is the speaker- and formant-intrinsic mean of all the vowel tokens analyzed for the speaker (all the vowels in the system are represented), and SD[Hz] is the speaker- and formant-intrinsic standard deviation. Our normalization procedure was based on vowel spaces calculated from multiple tokens of all three point vowels (/i a u/) obtained from these speakers. First and second formant values in Hertz were extracted from all these vowels in order to examine the vowel spaces of all individual participants. Ninety vowels per subject, 30 /i/, 30 /a/ and 30 /u/, for a total of 1,800 point vowels, including native and non-native productions by each participants, were recorded [Simonet, 2010a]. Twenty-two tokens were excluded due to recording errors or devoicing, thus resulting in a dataset of 1,778 vowel tokens. The speaker- and formant-

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intrinsic mean and standard deviation of both F1 and F2 of these data were calculated and used in the formula above.

Statistical Analyses

Linear mixed-effects regression was used for all statistical investigations. Mixed-effects regression takes fixed and random predictors. In all cases, random effects were: (1) random intercepts and slopes for subject, and (2) random intercepts for word. ANOVAs (on the χ^2 distribution, $p < 0.05$) performing pairwise comparisons of models with different definitions of random effects showed that models defined in this way were statistically not different from models with defined random intercepts and slopes for both subject and word, whereas they had a significantly better fit than models with random intercepts, but not slopes, for both subject and word. Mixed-effects models were conducted with the LME4 package [Bates and Maechler, 2010] of the computing language R. For these statistical models, we report the resulting estimate mean (β), its associated standard error (SE) and its corresponding t value. The significance (α) level was set at 5% ($p < 0.05$). In order to determine whether significance was reached we check whether $t < -2$ or $t > 2$. The gender of the speakers was not included as a predictor in the statistical models due to the fact that the speaker-intrinsic normalization procedure (see above) minimized these effects.

First, we compared Catalan and Spanish mid-back vowels as produced by dominant speakers of each language. In other words, a subset of the dataset comprising only the native productions of the 20 bilingual subjects was extracted. This subset was used to compare Catalan and Spanish mid-back vowels. For reasons explained above, /o/ (closed) and '/ɔ/' (open) tokens were included in both Catalan and Spanish. Therefore, structural, 'phonemic' effects of vowel (/o/-/ɔ/) were investigated in all statistical comparisons. In other words, the corresponding models had vowel (closed-open, within-subject) and language (Catalan-Spanish, between-subject) as fixed factors.

Second, we investigated Catalan as well as Spanish vowels as produced by dominant and nondominant speakers of the two languages. That is, we compared first the production of Catalan /o/ and /ɔ/ vowels by Catalan- and Spanish-dominant subjects. Then, we explored the production of Spanish /o/ and '/ɔ/' vowels by Spanish- and Catalan-dominant participants. Mixed-effects regression models with vowel (closed-open, within-subject) and native language (Catalan- vs. Spanish-dominant, between-subject) were used. These analyses addressed the following questions: Do speakers' productions in the same language differ as a function of their dominant language? Do Spanish-dominant subjects produce a Catalan /o/-/ɔ/ contrast? If they do, is their production of the contrast nativelike? That is, do their Catalan /o/ and /ɔ/ resemble those produced by Catalan-dominant participants? On the other hand, the analysis of Spanish vowels investigated whether Catalan-dominant speakers transfer a spurious native-language contrast to their nondominant language or whether they are nativelike in their Spanish productions.

Finally, the potential interactions of the vowels of the dominant and nondominant languages were analyzed in a series of within-speaker comparisons. Formant data were inspected in mixed-effects regression models as a function of vowel (closed-open, within-subject) and language (Catalan-Spanish, within-subject) as fixed factors. The questions addressed in these comparisons are as follows: Do early bilinguals maintain separate L1 and L2 systems? Do they, on the other hand, have merged L1 and L2 systems? Is there any interaction (i.e. assimilation or dissimilation) between L1 and L2 categories?

Results

Native Vowels: Catalan versus Spanish

In order to compare mid-back vowels in Catalan and Spanish, the Catalan productions of the Catalan-dominant subjects were examined alongside the Spanish productions of the Spanish-dominant subjects. Thus, only the native or dominant productions

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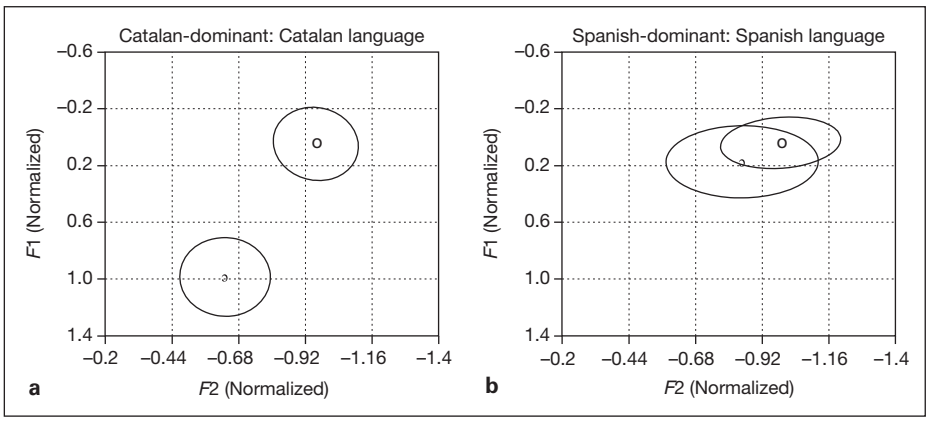

Fig. 1. Normalized F1 and F2 of native-produced Catalan and Spanish mid-back vowels as a function of vowel type (/o/-/ɔ/) and language (dominant speakers of each language, only).

of the 20 Spanish-Catalan bilinguals were investigated. It was expected for Catalan speakers to produce an acoustic contrast, in both height and fronting, between Catalan /o/ and /ɔ/. No such contrast was expected to occur in Spanish. Other questions were also addressed: Does Spanish /o/ occupy a space similar to Catalan /o/? Does it, on the other hand, overlap with both vowels? Or does it rather overlap with none of the two Catalan vowels? First, it could be hypothesized for Catalan /o/ to occupy the same acoustic (F1 × F2) space as Spanish /o/. Other possibilities, however, may be entertained. For instance, since Spanish /o/ exists in a less 'crowded' vowel system than Catalan /o/ and /ɔ/, the Spanish vowel could have a wider distribution than the Catalan vowels, thus partially (or completely) overlapping with both Catalan mid-back vowels.

Figure 1 displays two F1 × F2 charts plotting the Catalan and Spanish mid-back vowels produced by dominant speakers of each language as a function of vowel type (closed-open) and language (Catalan-Spanish). Inspection of these vowel plots suggests that Catalan-speaking individuals implement a Catalan /o/-/ɔ/ contrast in their speech that has no equivalent in Spanish. Apparently, the magnitude of the difference of the two Catalan mid-back vowels is greater on the F1 than on the F2 dimension.

Height (F1). A linear mixed-effects analysis with vowel type (closed-open) and language (Catalan-Spanish) yielded significant effects of vowel as well as a significant interaction between the two (table 2). The interaction was explored by analyzing the effects of vowel for each language separately. For Catalan, the model revealed robust vowel effects ($\beta = 0.94$, $t = 13.4$), while for Spanish no significant effects were found ($\beta = 0.13$, $t = 1.59$). Therefore, F1 values confirmed the existence of two height categories for Catalan but not for Spanish, as expected. A complementary investigation of the vowel-by-language interaction revealed that while Spanish /o/ did not differ from Catalan /o/ ($\beta = -0.008$, $t = -0.08$), Catalan /ɔ/ had a significantly higher F1 than Spanish '/ɔ/' ($\beta = -0.816$, $t = -7.04$). In other words, it was clear that Catalan /o/ occupied the same acoustic space, regarding vowel height, as Spanish /o/ and '/ɔ/' while Catalan /ɔ/ was significantly lower than the other three vowels.

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Table 2. Summaries of linear mixed-effects regression models fitted to native-produced Catalan and Spanish vowels with normalized F1 and F2 values as response(s), random intercepts for subject and word and random slopes for subject, and fixed effects for vowel (closed-open) and language (Catalan-Spanish)

Response	Predictor	Estimate	SE	t value
F1	(Intercept)	0.04		
	Open	0.94	0.07	12.18
	Spanish	–0.01	0.1	NS
	Open:Spanish	–0.81	0.1	–7.38
F2	(Intercept)	–0.96		
	Open	0.32	0.14	2.3
	Spanish	–0.03	0.14	NS
	Open:Spanish	–0.18	0.19	NS

If t value is <2 and >–2, then p = NS.

Fronting (F2). With respect to F2 data, we found moderately significant effects of vowel, but no effects of language and no significant interactions. Apparently, open vowels were slightly more fronted than closed vowels in both Spanish and Catalan. Since this was an unexpected result, we explored it further by dividing the dataset into two subsets as a function of language. It was found that Catalan significantly distinguished between /o/ and /ɔ/ in fronting ($\beta = 0.326$, $t = 3.31$), while Spanish did not distinguish between /o/ and '/ɔ/' ($\beta = 0.143$, $t = 0.82$). It would seem that only when pooling all the data together, i.e. Catalan and Spanish vowels, there would appear to be a contrast between /o/ and '/ɔ/' in Spanish. However, when explored further, this 'contrast' did not stand statistical testing. We believe that a mere tendency to differentiate between Spanish /o/ and '/ɔ/' for reasons not phonemic becomes a significant finding when these data are pooled together with those coming from a language with a phonemic /o/-/ɔ/ distinction. We conclude, therefore, that Catalan contrasts in fronting between /o/ and /ɔ/ while Spanish does not. A complementary model analyzing the effects of language for the two vowel types separately showed that Catalan /o/ was not different from Spanish /o/ ($\beta = –0.030$, $t = –0.23$) as well as Catalan /ɔ/ and Spanish '/ɔ/' did not differ from each other ($\beta = –0.213$, $t = –1.34$).

To summarize, the analyses reported in this section have revealed that the two phonemic Catalan mid-back vowels, /o/ and /ɔ/, are robustly different in both acoustic fronting and height. Spanish, which has only one mid-back vowel phoneme, also has only one mid-back vowel phonetic distribution. Spanish /o/ is different from Catalan /ɔ/ in height but is identical to Catalan /o/ in both height and fronting. Thus, Catalan /ɔ/ is located in an acoustic position not exploited by Spanish. That is, no overlap is evident between Catalan /ɔ/ and Spanish /o/. The following sections study the non-native productions of the Spanish-Catalan bilingual speakers investigated here.

Catalan Vowels

This section explores the Catalan productions of both Catalan-dominant and Spanish-dominant bilingual subjects. We address here the following research question:

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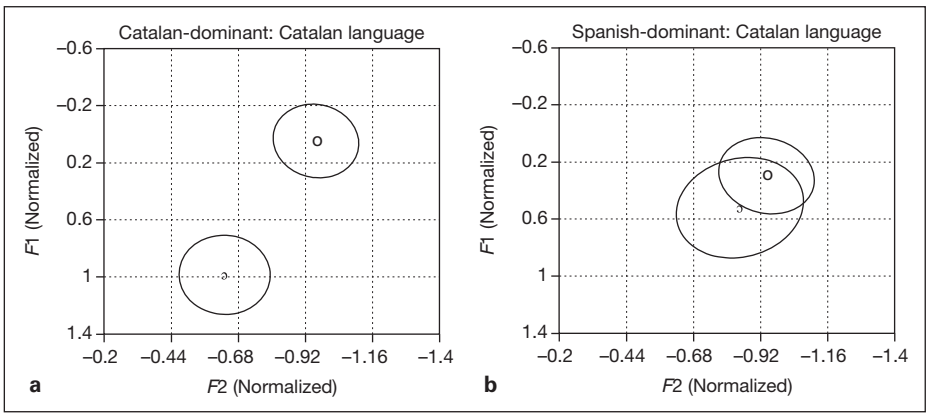

Fig. 2. Normalized F1 and F2 of Catalan mid-back vowels as a function of vowel type (/o/-/ɔ/) and native or dominant language group.

Do dominant and nondominant speakers of Catalan differ in their production of the Catalan /o/-/ɔ/ contrast? Since no such contrast exists in Spanish, we hypothesize that Spanish-dominant speakers of Catalan do not implement such a contrast in their nondominant language. If they do, however, are these productions native-like? That is, it seems possible that, although they might manage to produce a contrast between the two Catalan mid-back vowels, the exact acoustic (F1 × F2) locations of the two vowel categories might differ from those of the vowels produced by the Catalan-dominant subjects.

Figure 2 plots the Catalan /o/ and /ɔ/ vowels produced by the two groups of Spanish-Catalan bilinguals as a function of language dominance. The Catalan-dominant vowels have already been plotted in figure 1, but are here shown alongside nondominant Catalan productions for convenience. Cursory inspection of the vowel charts suggests that nondominant speakers of Catalan differ greatly from dominant speakers. Apparently, Spanish-dominant participants do not produce the Catalan /o/-/ɔ/ contrast in a nativelike manner.

Height (F1). The model analyzing normalized F1 data yielded significant vowel and dominant language effects, as well as a significant vowel by dominant language interaction (table 3). The two-way interaction was caused by the fact that the Catalan-dominant subjects produced a contrast between /o/ and /ɔ/ (see above) while the Spanish-dominant speakers did not ($\beta = 0.220$, $t = 1.77$). The interaction may also be seen in that dominant and nondominant tokens of Catalan /o/ differed from each other ($\beta = 0.249$, $t = 2.71$), and dominant productions of Catalan /ɔ/ were much lower than non-dominant productions of this vowel ($\beta = -0.469$, $t = -4.33$). This was also the cause of the dominant language fixed effect. In sum, Spanish-dominant speakers did not produce the Catalan /o/-/ɔ/ contrast in height. Their merged Catalan mid-back vowel was very different from Catalan /ɔ/ and very similar to (i.e., highly overlapping with) Catalan /o/ though still significantly different from it. It appears that the merged Catalan /o/ + /ɔ/ produced by the Spanish-dominant subjects is intermediate in height to the two native-produced mid-back vowels.

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Table 3. Summaries of linear mixed-effects regression models fitted to Catalan vowel productions with normalized F1 and F2 values as response(s), random intercepts for subject and word and random slopes for subject, and fixed effects for vowel (closed-open) and native language group (Catalan-dominant, Spanish-dominant)

Response	Predictor	Estimate	SE	t value
F1	(Intercept)	0.04		
	Open	0.94	0.09	9.72
	Spanish-dominant	0.24	0.09	2.71
	Open:Spanish-dominant	–0.71	0.08	–8.97
F2	(Intercept)	–0.95		
	Open	0.31	0.09	3.22
	Spanish-dominant	0.01	0.02	NS
	Open:Spanish-dominant	–0.23	0.04	–5.41

If t value is <2 and >–2, then p = NS.

Fronting (F2). The analysis of the normalized fronting values yielded significant vowel effects and a significant vowel-by-dominant language interaction. The interaction was due to the fact that the Spanish-dominant speakers of Catalan did not produce the Catalan /o/-/ɔ/ contrast in fronting ($\beta = 0.086$, $t = 0.81$) while the Catalan-dominant subjects did (see above). The two-way interaction was also triggered by the fact that dominant and nondominant productions differed for the Catalan open vowels ($\beta = -0.213$, $t = -4.45$) but not for the closed vowels ($\beta = 0.0172$, $t = 0.64$).

To summarize, dominant and nondominant Catalan productions differed in that dominant bilinguals produced a contrast between /o/ and /ɔ/ in both vowel height and fronting while nondominant speakers did not realize such a vowel contrast. Therefore, it may be concluded that Spanish-dominant speakers of Catalan have not, for the most part, learned to produce this Catalan-specific vowel contrast. Interestingly, however, we found that the single, merged Catalan mid-back vowel produced by the Spanish-dominant subjects was intermediate (on the F1 dimension) to the two native-produced Catalan vowels. While this merged non-native vowel differed from both native-produced vowels, it was more similar in height to Catalan-dominant /o/ than to Catalan-dominant /ɔ/.

Spanish Vowels

Do Catalan-dominant speakers differ from Spanish-dominant ones in their production of Spanish mid-back vowels? By hypothesis, Catalan-dominant subjects could impose a spurious /o/-/ɔ/ contrast in their Spanish at least in Spanish words that have Catalan cognates (our materials). Alternatively, Catalan-dominant subjects could differ from Spanish-dominant ones in their production of Spanish /o/ by producing an overall lower or more fronted vowel as a result of assimilation to their native, Catalan /ɔ/. A third possibility is that they do not differ from Spanish-dominant speakers.

Mixed-effects regression models of both normalized F1 and F2 reveal only effects of dominant language and only for F2 (table 4). Therefore, Spanish- and Catalan-dominant bilingual speakers did not produce a difference between Spanish /o/ vowels

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Table 4. Summaries of linear mixed-effects regression models fitted to Spanish vowel productions with normalized F1 and F2 values as response(s), random intercepts for subject and word and random slopes for subject, and fixed effects for vowel (closed-open) and native language group (Catalan-dominant, Spanish-dominant)

Response	Predictor	Estimate	SE	t value
F1	(Intercept)	0.03		
	Open	0.12	0.07	NS
	Spanish-dominant	0	0.07	NS
	Open:Spanish-dominant	0	0.02	NS
F2	(Intercept)	–0.9		
	Open	0.11	0.16	NS
	Spanish-dominant	–0.08	0.03	–2.58
	Open:Spanish-dominant	0.02	0.02	NS

If t value is <2 and >–2, then p = NS.

that have Catalan /o/ versus /ɔ/ cognates. On the other hand, notice that the Spanish /o/ produced by the Catalan-dominant subjects was slightly more fronted than that produced by Spanish-dominant speakers.

The following two sections report on a series of within-speaker comparisons that investigate whether bilinguals use different systems or vowel categories for their two languages or fundamentally transfer their native vowels to their non-native language.

Catalan-Dominant Speakers' Vowels

This section examines the Catalan and Spanish mid-back vowel productions of the Catalan-dominant bilingual speakers. In particular, we conduct a within-subject comparison of dominant and nondominant productions for these bilingual subjects. Do Catalan-dominant speakers transfer the characteristics of the vowels of their dominant to their nondominant language? It was shown above that these bilingual speakers do not transfer a spurious closed-'open' contrast to their Spanish. Furthermore, data reported in previous sections showed that their Spanish productions were native- or dominantlike except for the fact that their Spanish /o/ was slightly more fronted than the one produced by Spanish-dominant speakers. One hypothesis is that, since Spanish /o/ and Catalan /o/, as produced by dominant speakers, were identical (see above), the Catalan-dominant speakers might simply produce their Catalan /o/ as their Spanish /o/; that is, they might transfer an L1 category to their L2. Alternatively, since a difference in fronting was found between Spanish /o/ produced by dominant and nondominant speakers (see above), the Catalan-dominant subjects might have developed a separate Spanish /o/ category, different from both of their Catalan mid-back vowel categories. Figure 3 displays vowel charts as a function of vowel type and language.

Height (F1). The mixed-effects regression model fitted to normalized F1 had vowel (closed-open) and language (Catalan-Spanish) as fixed factors. Significant effects of vowel were found as well as a significant vowel-by-language interaction (table 5). The interaction was due to the fact that these bilingual speakers maintained a phonetic

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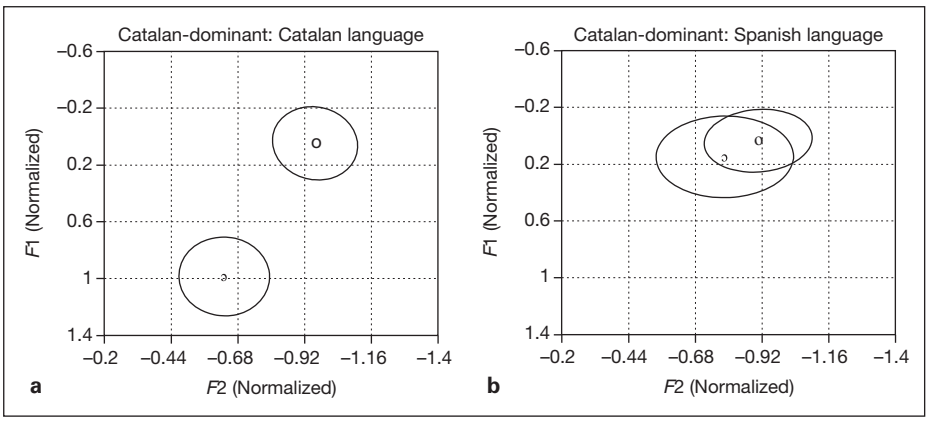

Fig. 3. Normalized F1 and F2 of mid-back vowels produced by Catalan-dominant bilinguals as a function of vowel type (/o/-/ɔ/) and language.

Table 5. Summaries of linear mixed-effects regression models fitted to Catalan and Spanish vowel productions by Catalan-dominant speakers with normalized F1 and F2 values as response(s), random intercepts for subject and word and random slopes for subject, and fixed effects for vowel (closed-open) and language (Catalan-Spanish)

Response	Predictor	Estimate	SE	t value
F1	(Intercept)	0.04		
	Open	0.94	0.06	14.13
	Spanish	–0.01	0.06	NS
	Open:Spanish	–0.85	0.08	–9.07
F2	(Intercept)	–0.95		
	Open	0.32	0.13	2.44
	Spanish	0.05	0.13	NS
	Open:Spanish	–0.21	0.18	NS

If t value is <2 and >–2, then p = NS.

contrast between Catalan /o/ and /ɔ/ (see above) while no equivalent contrast was evident between Spanish /o/ and '/ɔ/' (β = 0.129, t = 1.84). Furthermore, a significant difference was found between Catalan /ɔ/ and Spanish '/ɔ/' (β = –0.827, t = –10.96), while no such difference was evident between Catalan /o/ and Spanish /o/ (β = –0.011, t = –0.24).

Fronting (F2). A mixed-effects model with vowel (closed-open) and language (Catalan-Spanish) as predictors resulted in moderately significant vowel effects, but no other fixed effects nor interactions reached significance. Further explorations showed that, although there was no vowel-by-language interaction, Catalan mid-back vowels proved to be different for these subjects (see above) while Spanish mid-back vowels were not (β = 0.116, t = 0.71). Catalan /o/ did not differ from Spanish /o/ (β = 0.05, t = 0.42) and Catalan /ɔ/ did not differ from Spanish '/ɔ/' (β = –0.161, t = –1.11). In other words, there was an overall tendency to distinguish between closed and open vowels in

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fronting, but this tendency was more robust in Catalan than in Spanish. In fact, when dividing the two data sets as a function of language, the overall effect of vowel type was found to be solely due to the Catalan phonemic contrast and was not found in the Spanish cognates. Catalan and Spanish mid-back vowels did not differ from each other in fronting for these bilingual subjects.

To summarize, the Catalan-dominant speakers were found to produce a contrast between closed and open mid-back vowels in Catalan, their dominant language. No similar contrast was found in their production of the Spanish cognates. Therefore, they did not transfer a spurious contrast from their dominant into their nondominant language. It was found that these subjects' Spanish /o/ is not different from their native, Catalan /o/ while it differs greatly from their native, Catalan /ɔ/. Therefore, the Catalan-dominant speakers seem to have transferred their Catalan /o/ to their Spanish. They may be claimed to posses a two-category mid-back vowel system: (1) one vowel for both Catalan /o/ and Spanish /o/, and (2) another one for Catalan /ɔ/.

Spanish-Dominant Speakers' Vowels

This section explores the mid-back vowel (sub)systems of the Spanish-dominant bilingual participants. The main question we address is as follows: Have Spanish-dominant speakers been able to learn the Catalan /o/-/ɔ/ contrast and do they implement it in their speech? Data reported in the previous sections have shown that this was not so. However, other effects were suggested in previous comparisons that we explore here in detail. For instance, it was found that the Spanish /o/ produced by these speakers did not differ from native-produced Catalan /o/ (see above). On the other hand, the Catalan /o/ produced by these speakers differed from native-produced Catalan /o/ in vowel height (see above). This suggests that the Spanish-dominant bilingual speakers might have developed a lower (with respect to their native Spanish /o/) mid-back, merged vowel category for Catalan. The direct within-subject analyses reported here investigate this hypothesis.

Height (F1). Normalized F1 data were explored with a main mixed-effects model with vowel (closed-open) and language (Catalan-Spanish) as fixed factors. The model showed marginal effects of vowel as well as significant effects of language, but no interaction between the two (table 6, fig. 4). The marginal effects of vowel do not stand further statistical scrutiny when the data are divided into two subsets as a function of language. Thus, the Spanish-dominant subjects did not maintain an acoustic difference between Catalan /o/ and /ɔ/ ($\beta = 0.22$, $t = 1.77$) as they did not maintain a difference between their corresponding Spanish cognates (see above). The fixed effects of language are of interest. Evidently, the merged Catalan /o/ + /ɔ/ vowel produced by these bilinguals is lower (i.e., has a higher F1) than their Spanish /o/.

Fronting (F2). Normalized F2 values were submitted to identical statistical explorations; that is, we used mixed-effects modeling with vowel (closed-open) and language (Catalan-Spanish) as fixed predictors. No significant effects and no interactions were found for these vowels in their fronting dimension.

In summary, Spanish-dominant speakers tended not to reliably produce the /o/-/ɔ/ contrast in their nondominant language, Catalan. An interesting finding reported in this section was that the Spanish-dominant subjects produced a lower (merged) mid-back vowel in Catalan than in Spanish. Therefore, Spanish-dominant speakers seem to have

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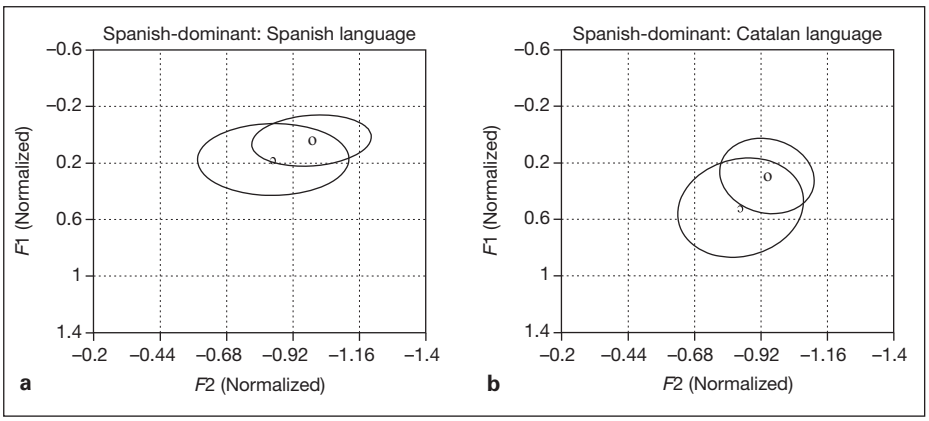

Fig. 4. Normalized F1 and F2 of mid-back vowels produced by Spanish-dominant subjects as a function of vowel type (/o/-/ɔ/) and language.

Table 6. Summaries of linear mixed-effects regression models fitted to Catalan and Spanish vowel productions by Spanish-dominant speakers with normalized F1 and F2 values as response(s), random intercepts for subject and word and random slopes for subject, and fixed effects for vowel (closed-open) and language (Catalan-Spanish)

Response	Predictor	Estimate	SE	t value
F1	(Intercept)	0.29		
	Open	0.22	0.1	2.18
	Spanish	–0.25	0.09	–2.62
	Open:Spanish	–0.08	0.13	NS
F2	(Intercept)	–0.94		
	Open	0.08	0.13	NS
	Spanish	–0.04	0.14	NS
	Open:Spanish	0.06	0.19	NS

If t value is <2 and >–2, then p = NS.

developed a 'new' phonetic category for their nondominant language. As reported in previous sections, this 'new' Catalan vowel was unlike any of the two native-produced Catalan vowels (/o/-/ɔ/). In fact, this 'new' vowel category was located, on the F1 dimension, between Catalan /o/ and /ɔ/. That is, the Spanish-dominant bilingual speakers investigated here were found to possess a two-category mid-back vowel system: (1) one for their Spanish /o/ and (2) another one for their Catalan merged /o/ + /ɔ/.

Discussion

This study has explored the acoustics of Spanish and Catalan mid-back vowels as produced by Catalan- and Spanish-dominant bilinguals residing on Majorca. Native as

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well as non-native productions of both languages were investigated. Catalan has two phonemic mid-back vowels: /o/ and /ɔ/. Spanish, on the other hand, has only one: /o/. The results of the analyses reported in this study may be summarized as follows:

First, Spanish /o/ and Catalan /o/ and /ɔ/ were compared; that is, we compared the acoustic characteristics of these vowels as produced by native, L1-dominant speakers of the two languages. As expected, Catalan /ɔ/ was found to be lower and more fronted than Catalan /o/. A difference was also evident between Catalan /ɔ/ and Spanish /o/. On the other hand, Catalan /o/ and Spanish /o/ were found to be acoustically indistinguishable.

Second, a comparison of dominant and nondominant productions of Catalan /o/ and /ɔ/ revealed that Spanish-dominant subjects did not produce the Catalan-specific vowel contrast whereas Catalan-dominant participants did. That is, Spanish-dominant subjects produced a single, merged phonetic category (acoustic distribution) instead of two. Third, an analysis of dominant and nondominant productions of Spanish /o/ did not reveal any differences between Catalan- and Spanish-dominant participants. Catalan-dominant speakers did not produce a spurious contrast in their nondominant language. Since dominant productions of Catalan /o/ and Spanish /o/ had not shown any acoustic difference between these two vowels, it is hypothesized that Catalan-dominant subjects simply used their Catalan /o/ to produce their Spanish /o/.

Finally, we carried out a series of within-speaker analyses in which we compared the vowels of the dominant and nondominant languages of the bilinguals. On the one hand, the Catalan-dominant speakers were found to have a system of two mid-back vowel categories, one for Catalan /ɔ/ and another one for Catalan /o/ and Spanish /o/. Thus, no differences were found between Catalan /o/ and Spanish /o/ for these subjects. On the other hand, the findings with respect to the productions of the Spanish-dominant speakers were rather unexpected. First, they did not produce a difference between Catalan /o/ and Catalan /ɔ/, as mentioned above. In other words, they produced a merged phonetic category for the two Catalan mid-back vowels. The unexpected finding, however, was that they had two different acoustic distributions in their productions: one for Spanish /o/ and one for Catalan /o/ and /ɔ/. Their Catalan mid-back vowel category was significantly lower than their Spanish mid-back vowel category. These findings have a number of implications.

Consider first the finding that Spanish-dominant bilinguals did not produce an acoustic difference between Catalan /o/ and /ɔ/. While some studies have shown that early bilinguals might be able to develop native-like pronunciation patterns for their non-native language [e.g., Flege et al., 1999a, b; Munro et al., 1996], others have reported differences between the productions of early bilinguals and native speakers of the L2 [e.g., Guion et al., 2000; Mack, 1989; Piske et al., 2002]. The present study has found that Spanish-dominant speakers are not able to produce Catalan mid-back vowels in a native-like manner. These results therefore appear to support the hypothesis that 'severe limitations to the malleability of the initially acquired L1 phonemic categories [persist], even under conditions of early and extensive exposure' [Sebastián-Gallés and Soto-Faraco, 1999, p. 120]. However, notice that the bilinguals whose speech was investigated in the present study were frequent users of their native language; that is, the language they were first exposed to was also the language they continued to use more frequently. It is expected for the frequencies of use of the two languages of bilinguals to be negatively correlated; that is, the more a bilingual uses his L1, the less he uses his L2 and vice versa. Effects of L1-L2 use

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have been reported in a number of papers. For example, Flege et al. [1997] asked native English listeners to rate the degree of 'foreign' accent of English sentences produced by two groups of early Italian-English bilinguals (i.e., two groups of speakers who migrated to North America during childhood) differing in Italian language use. Both groups were found to have some degree of 'foreign' accent in their speech; that is, they both differed from native speakers of English. However, the bilinguals who used Italian seldom had a milder 'foreign' accent than those who used Italian frequently, thus showing that continuing to use a native language frequently might negatively impact L2 speech learning. Other studies have reported similar findings. On the one hand, Guion et al. [2000] and Piske et al. [2002] found audible differences between the productions of early bilinguals who use their native language frequently (and their non-native language seldom) and native speakers of the second language. On the other hand, these studies did not find differences between the productions of early bilinguals who used their L1 seldom and L2 native speakers. Returning to our study, it is possible that the differences between Spanish- and Catalan-dominant bilinguals with respect to the production of Catalan /o/ and /ɔ/ reported here are due not only to a difference in the order of acquisition of the two languages but also to the different patterns or frequencies of use of the two languages. Based on the evidence reviewed here, we conclude that 'early and extensive exposure' does not guarantee that learners will be able to acquire the speech patterns of their L2 in a nativelike manner [Sebastián-Gallés and Soto-Faraco, 1999, p. 120]. We use the term 'extensive' here to refer to a kind of exposure that occurs throughout the life span of a bilingual, i.e. that systematically persists since the time of first L2 exposure. On the other hand, we do not know how 'intensive' their L2 exposure was, i.e. how much interaction in their L2 they had with L2-dominant speakers. In fact, although it is key to determine the intensity of L2 exposure, for a complex sociolinguistic situation like that in Majorca, it is extremely difficult to assess this. In sum, although the available evidence suggests that sequential learning, even if it occurs during childhood, has persistent, lifelong effects, we may not rule out at present the possibility for early bilinguals who become frequent users of their L2 (and who seldom use their L1) to learn to produce L2 sounds in a nativelike manner.

Let us now consider the unexpected finding that Spanish-dominant speakers showed evidence for a two-category mid-back vowel system, i.e. one Spanish and one Catalan mid-back vowel. This finding is particularly surprising if we take into account the acoustics of Spanish /o/ and Catalan /o/ as produced by native speakers of the two languages. Note that it was found that the two vowels are acoustically indistinguishable. Therefore, it was expected that Spanish-dominant subjects would produce a Catalan /o/ that would not differ from that produced by Catalan-dominant participants. Differences were perhaps expected between these two groups with respect to Catalan /ɔ/, which is the only one that does not have an acoustic equivalent in Spanish. Since we found that Spanish-dominant speakers merged the two Catalan mid-back vowels into a single category (hypothetically due to L1-L2 perceptual assimilation patterns), we initially expected that they would perhaps use their Spanish /o/ for both their Catalan /o/ and /ɔ/. However, it was found that the merged Catalan /o/ + /ɔ/ vowel produced by the Spanish-dominant subjects differed from both their own Spanish /o/ and the two Catalan mid-back vowels produced by native speakers of Catalan.

The fact that the Spanish-dominant bilinguals had developed a 'new' phonetic category to accommodate an L2 vowel provides evidence of 'learning' for these subjects.

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This is so because it is known that only early, highly proficient bilinguals are generally able to develop separate phonetic categories for the sounds of their L2 and this is generally interpreted as evidence of L2 speech development or learning [Antoniou et al., 2010; Flege, 2007; Fowler et al., 2008; Guion, 2003; Kang and Guion, 2006; Piske et al., 2002]. For instance, Guion [2003] found that some simultaneous and early Quichua-Spanish bilinguals produced separate acoustic categories for Quichua /ɪ/ and Spanish /i/ and /e/, Quichua /u/ and Spanish /u/ and /o/, and Quichua /a/ and Spanish /a/. Late, inexperienced Quichua-Spanish bilinguals, on the other hand, tended to produce Quichua /ɪ/ and Spanish /i/ and /e/ as /ɪ/, Quichua /u/ and Spanish /u/ and /o/ as /u/, and, finally, Quichua and Spanish /a/ as one single vowel category. However, for the Spanish-Catalan bilinguals investigated here, 'learning' (the development of a separate, L2 category) did not imply approximation to native speaker productions but rather a departure from them. Why did Spanish-dominant subjects develop a Catalan mid-back vowel that did not resemble any of the two Catalan mid-back vowels? Let us consider some recent findings (and theoretical models) that may shed some light on the issue.

First, recent investigations have shown that the phonetic characteristics of L2 sounds are able to impact L2 speech learning in cases where there exist phonetic differences between a single L1 sound and a single L2 sound, even when no phonological, systemic differences exist between the two languages. For instance, Tsukada et al. [2004] explored the production of release bursts in word-final stop consonants by Korean learners of English. While English variably releases word-final stop consonants, Korean equivalents are invariably unreleased. Korean learners were recorded twice, with a difference of 1 year between the first and the second recordings. These authors showed evidence of an increase in the release of word-final stops for the learners. This indicated that production learning is possible even in cases where there is a one-to-one L1-L2 correspondence. Similarly, MacKay et al. [2001] analyzed the productions of English /b/ by Italian learners of English. While Italian /b/ is invariably prevoiced, English /b/ variably presents prevoicing or a short-lag voice onset time (VOT). It was found that the Italian-English bilinguals prevoiced their English /b/ more than English controls but less than Italian controls. This suggested that even low-level phonetic differences that are not used to signal L2-specific contrasts may impact L2 speech learning. Returning to our study, we hypothesize that the Catalan input (presence of /o/ and /ɔ/) received by Spanish-dominant bilinguals from Catalan-dominant subjects may have had an effect on the Catalan productions of the Spanish-dominant speakers. Although this did not lead to the development of two separate categories for Catalan /o/ and /ɔ/, the presence of lower mid-back (/ɔ/) vowel tokens in Catalan than in Spanish may have led Spanish-dominant bilinguals to posit a lower mid-back vowel for their Catalan than for their Spanish productions.

Second, some studies have presented evidence for a process known as phonetic category dissimilation [Flege, 1995, 2002, 2007]. Flege's [1995] Speech Learning Model (SLM) assumes that the phonetic systems of the two languages of a learner coexist in a common phonetic space. On the one hand, the model hypothesizes that if a particular sound of the L2 is assimilated (i.e., it is phonetically similar and thus classified as being 'equivalent') to a sound of the L1, learners will experience difficulties establishing a separate phonetic category for the L2 sound. If assimilation persists, a merged L1 + L2 phonetic category may develop. The merged category may display some features of the two assimilated sounds. Eventually, especially for early learners,

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assimilation might be overcome by some bilinguals and they might develop a separate category for the L2 sound. The new, separate category may or may not be native-like [Antoniou et al., 2010; Fowler et al., 2008].

When a separate L2 sound category has been developed by the bilingual learner, this results in a more 'crowded' phonetic space, according to the SLM [Flege, 1995]. Crowdedness may lead to dissimilation; that is, the phonetic distance between L1 and L2 sounds may be exaggerated (i.e., increased with respect to the distance between these two sounds as produced by monolingual speakers of the two languages) in order to accommodate the fact that they have developed a separate L2 category. For instance, Flege et al. [2003] analyzed the production of English /eɪ/ by early and late Italian learners of English. English /eɪ/ is assimilated to Italian /e/ by native Italians, although the two vowels differ in amount of tongue movement or formant dynamics, i.e. Italian /e/ is a monophthong while English /eɪ/ is a diphthongized vowel. It was found that late learners tended to produce English /eɪ/ with less formant change than English controls whereas early bilinguals produced it with more formant change than English controls. This was interpreted to show that the early Italian-English bilinguals, who had developed a separate category for English /eɪ/ (i.e., separate from Italian /e/), exaggerated the formant movement of this vowel to distinguish it from their own Italian /e/. Flege and Eefting [1987] also showed evidence of L1-L2 phonetic dissimilation. Two groups of Spanish-English bilinguals were found to produce Spanish /p t k/ with shorter VOTs than Spanish monolinguals. Since the bilinguals were shown to have established separate phonetic categories for English /p t k/, the authors interpreted these findings to show that their Spanish stops were dissimilating from their English equivalents. On the other hand, numerous studies have failed to find evidence for the operation of L1-L2 phonetic dissimilation [Antoniou et al., 2010; Flege, 1987, 1991; Fowler et al., 2008; Sancier and Fowler, 1997, among others]; that is, it is common for the finding to be that, even for subjects who have developed a separate L2 phonetic category, the 'new' category is intermediate between the L1 and the target L2 sounds. For instance, Sancier and Fowler [1997] found that one proficient late bilingual modified the duration of VOT in voiceless stops in the same direction in both her L1 and L2 after a stay in an L1/L2-speaking environment. Importantly, this proficient bilingual produced separate VOT distributions for her L1 and her L2, thus providing evidence of new category development. Returning to our study, it is possible that the difference between the Catalan and the Spanish mid-back vowels produced by these subjects arose as a consequence of the operation of L1-L2 category dissimilation. This finding is significant mostly due to the fact that the operation of category dissimilation, although predicted by Flege's [1995] SLM, is seldom observed (see above). In sum, the unpredicted L1-L2 acoustic difference reported here may have resulted from the combined operation of two processes: (1) the impact of the Catalan /ɔ/ input received, which may have 'encouraged' learners to produce a lower mid-back vowel in Catalan than in Spanish, and (2) L1-L2 phonetic category dissimilation between Spanish /o/ and the merged Catalan /o/ + /ɔ/ of these bilinguals.

Conclusion

Catalan has two mid-back vowels (/o/ and /ɔ/) while Spanish has only one (/o/). The present study investigated the mid-back vowel productions of two groups of

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Spanish-Catalan bilinguals residing on Majorca, where the languages are co-official. First, we found that Spanish /o/ and Catalan /o/ are virtually indistinguishable in their height and fronting, as evidenced by F1 and F2 values. Second, we showed that Catalan /ɔ/ is significantly lower and more fronted than the other two vowels. After exploring the acoustics of Catalan and Spanish mid-back vowels as produced by native speakers of both languages, we proceeded to investigate non-native vowel productions.

On the one hand, it was shown that Catalan- and Spanish-dominant bilinguals did not differ in their production of Spanish /o/; that is, Catalan-dominant speakers produced native-like Spanish /o/. On the other hand, the two groups of bilinguals differed in their production of Catalan /o/ and /ɔ/. First, it was found that the Spanish-dominant bilinguals did not produce the Catalan-specific /o/-/ɔ/ vowel contrast: they produced a single (merged) mid-back vowel. Second, we found that Spanish-dominant subjects produced two different mid-back vowel categories: one for their Spanish /o/ and one for their merged (/o/ + /ɔ/) Catalan equivalent. Interestingly, the Catalan mid-back vowel produced by the nondominant bilinguals differed from both the Catalan /o/ and /ɔ/ produced by Catalan-dominant subjects and from their own Spanish /o/. It has been proposed that the development of the two-category system of the Spanish-dominant subjects results from the operation of two processes: (1) the impact of the statistical, phonetic characteristics of Catalan /ɔ/ on the learning of Spanish-dominant subjects, and (2) phonetic category dissimilation, as operationalized in Flege's [1995, 2002, 2007; Flege and Eefting, 1987; Flege et al., 2003] SLM.

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